

Project Workflow: Week 7

Phase: Multiple Levels Implementation

1. Objective:

The goal of Week 7 is to expand the gameplay experience by introducing a multi-level structure with unique layouts, progressive difficulty scaling, and clean transition mechanics. This includes:

- Designing and implementing multiple distinct levels featuring varying obstacle layouts and road configurations.
- Adding dynamic level progression and difficulty scaling based on distance or score milestones.
- Implementing interactive level transition screens between stages.

2. Detailed Task Breakdown

- **Designing Multiple Level Configurations:** Established a structured framework for different stages (e.g., Level 1, Level 2, Level 3), customizing parameters such as enemy car spawn frequency, road curve intensity, and environmental density for each level.
- **Implementing Level Progression & Difficulty Scaling:** Developed a tracking mechanism that monitors player progress (distance traveled or score), dynamically adjusting parameters like base speed, obstacle velocity, and item appearance rates as players advance.
- **Building Level Transition Screens:** Created modal UI overlays or state-managed transition screens that trigger upon completing a level objective, displaying performance statistics and offering options to proceed to the next stage or restart.

3. Deliverables

- **Level Configurations:** Distinct stage layouts managed through a centralized level controller.
- **Difficulty Scaling System:** Dynamic adjustments to speed, spawn rates, and hazard density based on progress.

- **Transition Screens:** UI screens handling state switches between levels with completion stats.

4. Tools & Resources

- **Development Environment:** Java JDK 8+, IntelliJ IDEA / VS Code.
- **Reference Materials:** Java 2D Graphics UI rendering, state-machine patterns for game flow control, and event-driven screen transitions.